

PCOS Diagnosis

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[PCOS_diagnosis_ML](#) - code

Introduction

Polycystic ovary syndrome (**PCOS**) is a common hormonal disorder that affects women during their reproductive years and beyond, into later stages of life.

PCOS is a leading cause of irregular menstrual periods and one of the most common causes of **infertility**. Irregular or infrequent menstrual periods among women with PCOS may also raise their risk for **endometrial hyperplasia** and/or **endometrial cancer**.

Further, PCOS is a chronic metabolic condition associated with heightened long-term risk for **insulin resistance**, type **2 diabetes** mellitus and obesity.

PCOS can start as early as during adolescence but is most commonly detected or diagnosed when women struggle to become pregnant.

Therefore, In this report I will go into detail about each step of the project, from preparing the data and creating features to using three models for diagnosis.

Dataset Overview

The [dataset](#), sourced from Kaggle, contains medical, demographic, and hormonal information about patients.

It consists of:

- 541 patients records,
- 42 attributes, including:
 - Age
 - Body Mass Index (BMI)
 - FSH (Follicle-Stimulating Hormone)
 - Patient symptoms
 - Other clinical and laboratory measurements

The target variable indicates whether a patient is diagnosed with PCOS (binary classification: 0 = No PCOS, 1 = PCOS).

The dataset represents structured tabular medical data, making it suitable for classical machine learning models as well as neural networks.

Data Preprocessing

Before training the models, data needs to be prepared.

Feature Grouping

To better understand how different types of clinical information contribute to PCOS diagnosis, the dataset features were divided into three main groups:

1. Hormonal Features

Laboratory hormone measurements related to reproductive and metabolic function, including FSH, LH, FSH/LH ratio, AMH, TSH, prolactin, and progesterone levels.

2. Patient-Observable Features

Demographic information and self-reported symptoms such as age, weight, menstrual cycle regularity, lifestyle habits (fast food, exercise), and visible symptoms like hair growth, acne, or hair loss.

3. Doctor-Observable Features

A set combining:

- Patient-Observable Features - Patient-reported symptoms
- Clinically measured indicators (BMI, blood pressure, waist–hip ratio, pulse rate, etc.)
- Doctor check up findings (Ultrasound findings (follicle count, follicle size), endometrium thickness...)

4. All Features

All features except identification columns (Sl. No and Patient File No.) and target value (PCOS (Y/N))

Handling Missing Values

Some features contained missing values. To ensure reliable results, missing values were removed.

Feature Scaling

Some medical features, such as hormone levels and BMI, are measured on very different scales. To prevent large values from dominating the model, numerical features were standardized using a scaling method.

Train–Test Split

The dataset was divided into training and testing sets.

- The training set was used to train the models.
- The test set was used to evaluate model performance on unseen data.

This helps measure how well the model generalizes to new patients.

Modeling Algorithms

To evaluate different approaches for PCOS diagnosis, three classification models were implemented:

Logistic Regression

Logistic Regression serves as a baseline model. It is a linear binary classifier. It calculates the probability of PCOS presence using a logistic (sigmoid) function.

Because it is simple and easy to intercept, it is useful for comparing results with more complex models.

Random Forest Classifier

Random Forest is a more advanced model that uses many decision trees to improve prediction. Each tree makes a prediction, and the final result is based on the majority vote of all trees.

Artificial Neural Networks

A feedforward Artificial Neural Network was implemented to model more complex patterns in the data. Neural networks can automatically learn nonlinear feature interactions and hidden structures that traditional models may not capture. The ANN allows comparison between classical machine learning approaches and deep learning methods applied to tabular medical datasets.

Model Evaluation

To evaluate the metric of the models, the several metrics were used:

- Accuracy - how many predictions model got right out of the all predictions
- Precision - focuses on the quality of the positive predictions
- Recall - measures how good the model is at predicting positives
- F1-score - balance between precision and recall
- Confusion matrix - shows True Positives, True Negatives, False Positives, and False Negatives

A False Negative means that a patient who actually has PCOS is predicted as healthy. This can delay diagnosis and treatment and may lead to serious health complications.

For this reason, during model selection and hyperparameter tuning, **recall was used as the main scoring metric**. The models were fine-tuned to maximize recall in order to minimize the number of False Negatives. In this context, it is more acceptable to have a False Positive (predicting PCOS when the patient is healthy) than a False Negative. A False Positive may lead to additional medical tests, but a False Negative may leave the condition untreated.

Therefore, the final models were selected not only based on overall accuracy, but mainly based on their ability to correctly identify PCOS patients.

Results and Discussion

Feature_Set	Model	Accuracy	Recall_1	Precision_1	F1_1	TN	FP	FN	TP
doctor_observable_features	Logistic Regression	0.8519	0.9565	0.7097	0.8148	71	18	2	44
doctor_observable_features	Random Forest	0.9259	0.9565	0.8462	0.898	81	8	2	44
doctor_observable_features	ANN	0.8593	0.8043	0.7872	0.7957	79	10	9	37
hormone_features	Logistic Regression	0.4667	0.7556	0.3579	0.4857	29	61	11	34
hormone_features	Random Forest	0.6889	0.6222	0.5283	0.5714	65	25	17	28
hormone_features	ANN	0.3926	0.9111	0.3445	0.5	12	78	4	41
patient_observable_features	Logistic Regression	0.8148	0.8043	0.6981	0.7475	73	16	9	37
patient_observable_features	Random Forest	0.8148	0.6957	0.7442	0.7191	78	11	14	32
patient_observable_features	ANN	0.837	0.7826	0.75	0.766	77	12	10	36
all_features	Logistic Regression	0.837	0.8421	0.6667	0.7442	81	16	6	32
all_features	Random Forest	0.8889	0.7895	0.8108	0.8	90	7	8	30
all_features	ANN	0.8519	0.7632	0.725	0.7436	86	11	9	29

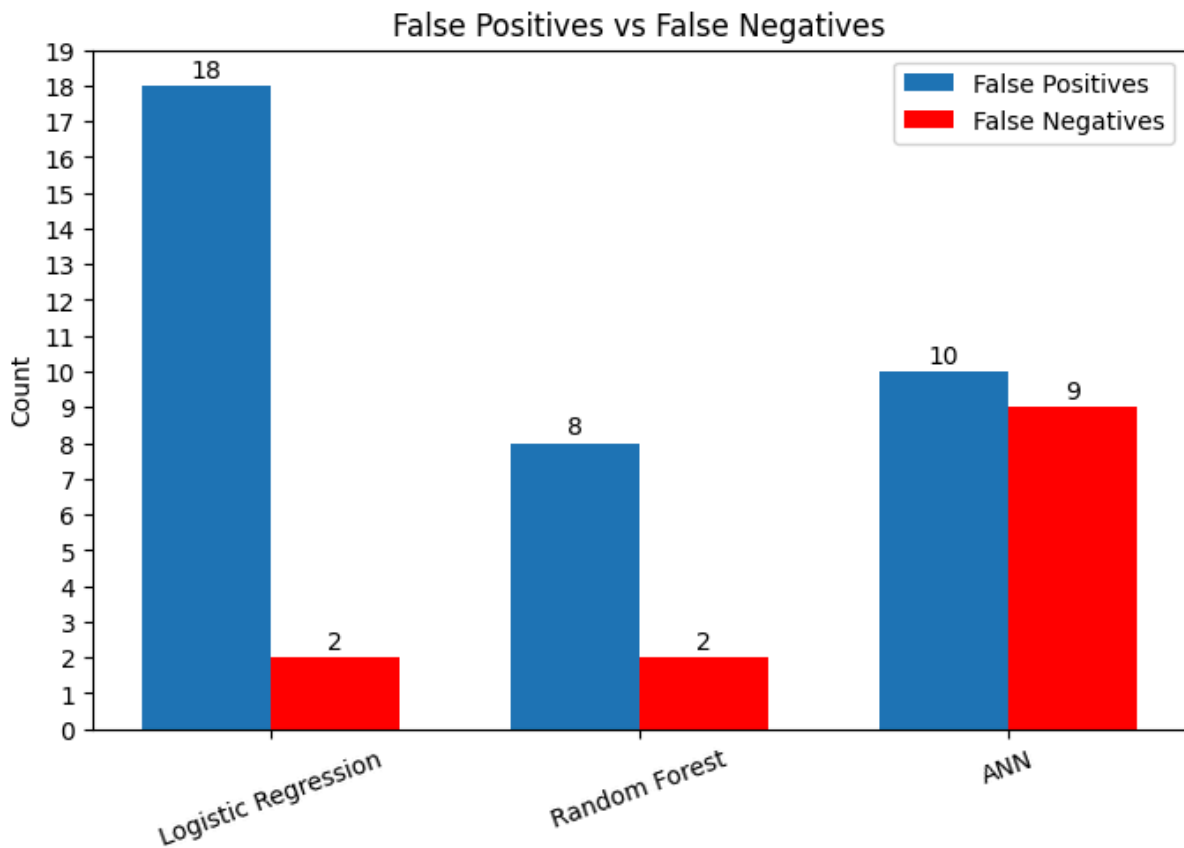
Among all models, Random Forest achieved the best overall performance when using **doctor observable features**. It reached accuracy of 92.59% and recall of 0.956, while balancing precision as well 0.846 and the highest f1 score 0.898. Additionally, it had 2 false negatives and 8 false positives.

Logistic Regression achieved the same high recall (0.956), also resulting in only two false negatives. This makes it particularly suitable in clinical scenarios where minimizing missed PCOS diagnoses is the main objective. However, it produced a larger number of false positives (18), which may lead to unnecessary follow-up examinations and increased clinical workload.

The Artificial Neural Network achieved competitive accuracy (85.93%) and a recall of 0.804. However, it resulted in slightly more false negatives (9) compared to the other models. This outcome is consistent with expectations, as neural networks typically require larger datasets to fully capture complex patterns and generalize effectively. With a relatively small tabular dataset, tree-based methods often outperform deep learning models.

Doctor Observable Features consistently outperformed both single-category features (hormonal or patient-reported) and the full feature set. This is likely because they combine high-quality patient symptoms with clinical measurements, providing the model with the most informative and reliable signals while avoiding the noise introduced by less relevant features

Model Comparison for Doctor Observable Features				
Model	Accuracy	Recall	Precision	F1
Logistic Regression	0.8519	0.9565	0.7097	0.8148
Random Forest	0.9259	0.9565	0.8462	0.898
ANN	0.8593	0.8043	0.7872	0.7957



Conclusion

This project demonstrated how machine learning models can assist in diagnosing Polycystic Ovary Syndrome using structured medical data.

In this study, we compared Logistic Regression, Random Forest, and ANN models for PCOS detection. Random Forest provided the best trade-off between accuracy, recall, and precision, making it the most suitable model for this dataset. Logistic Regression is a reliable option when maximizing recall is prioritized, but it produces a larger number of false positives. Artificial Neural Networks, while powerful in larger datasets and high-dimensional settings, did not outperform tree-based models in this study due to the limited dataset size and increased risk of overfitting.